

Infron vs OpenRouter Routing, Cache, Cost, and Streaming Performance A/B Test Report

Abstract and Executive Outline

Keywords: Prompt Caching; A/B Testing; Provider Routing; Cache Affinity; Latency; Throughput; Cost Attribution; qwen3.6-35b-a3b

Abstract

This report evaluates `qwen/qwen3.6-35b-a3b` on Infron and OpenRouter across cache reuse, observed cost, throughput, E2E latency, and Streaming TTFT under Prompt Caching workloads.

The main findings are: Infron and OpenRouter tie on Cache hit rate in all routing modes; Infron leads Observed cost in 1/4 routing modes; OpenRouter leads in 3/4; 0/4 tie; Infron leads Throughput in 1/4 routing modes; OpenRouter leads in 3/4; 0/4 tie; Infron leads E2E latency in all routing modes; Infron leads Streaming TTFT in 3/4 routing modes; OpenRouter leads in 1/4; 0/4 tie.

Overall, Infron's cross-mode strengths are Streaming TTFT, E2E latency, while OpenRouter's cross-mode strengths are throughput, observed cost. Platform choice should be driven by workload objective rather than a single headline metric.

Figure 0: Normalized Capability Radar

The five radar axes represent throughput, price, E2E latency, Streaming TTFT, and cache hit rate. Every metric is normalized to a 0–100 score, with farther outward meaning better.

Thick solid lines show platform–level contours; translucent lines and points show individual routing modes.

Conclusion Overview: Core Metric Winners by Routing Mode

Based on strict A/B paired samples. Blue represents Infron, orange represents OpenRouter; gold cells mark the winner for the routing objective.

Throughput OpenRouter wins 3/4 Max advantage 59.66%, higher is better	Observed cost OpenRouter wins 3/4 Max advantage 39.08%, lower is better	E2E latency Infron wins 4/4 Max advantage 79.79%, lower is better	Streaming TTFT Infron wins 3/4 Max advantage 40.81%, lower is better	Cache hit rate Tie in 4/4 modes Max advantage 0.00%, higher is better
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Routing mode	Throughput objective	Cost objective	Latency objective	TTFT objective	Cache result
Throughput First throughput	OpenRouter advantage 59.66%	Infron advantage 39.08%	Infron advantage 79.79%	OpenRouter advantage 40.81%	Tie tie
Price First price	OpenRouter advantage 8.40%	OpenRouter advantage 6.98%	Infron advantage 14.62%	Infron advantage 14.31%	Tie tie
Latency First latency	OpenRouter advantage 10.56%	OpenRouter advantage 6.96%	Infron advantage 12.78%	Infron advantage 13.51%	Tie tie
TTFT First ttft	Infron advantage 3.42%	OpenRouter advantage 6.96%	Infron advantage 28.90%	Infron advantage 25.80%	Tie tie

Executive Outline

Dimension	Conclusion	Evidence
Controls	First/second <code>usage.prompt_tokens</code> deltas are limited to 50 tokens within each <code>sort/group/round</code> pair.	Methods and data quality
Cache reuse	Infron and OpenRouter tie on Cache hit rate in all routing modes	Overall metrics and mechanism section
Observed cost	Infron leads Observed cost in 1/4 routing modes; OpenRouter leads in 3/4; 0/4 tie	Overall metrics and provider drill-down
Performance	Infron leads Throughput in 1/4 routing modes; OpenRouter leads in 3/4; 0/4 tie; Infron leads E2E latency in all routing modes; Infron leads Streaming TTFT in 3/4 routing modes; OpenRouter leads in 1/4; 0/4 tie	Charts and statistical tests
Attribution boundary	Claims use observable response telemetry: usage, cost, TTFT, latency, provider fields, and cache tokens.	Provider/Route drill-down
Business meaning	Long-context, RAG-prefix, agent-tool, and high-frequency template workloads should evaluate cache rate, cost, first-token latency, and E2E latency together.	Discussion and conclusion

Routing–Mode Conclusions

Routing mode	Objective	Cache winner	Cost winner	Throughput winner	E2E latency winner	TTFT winner	Interpretation
Throughput First	Maximize output capacity per unit time	Tie	Infron	OpenRouter	Infron	OpenRouter	Mixed result
Price First	Minimize request and token cost	Tie	OpenRouter	OpenRouter	Infron	Infron	Mixed result
Latency First	Minimize full–response waiting time	Tie	OpenRouter	OpenRouter	Infron	Infron	Mixed result
TTFT First	Minimize streaming first–token time	Tie	OpenRouter	Infron	Infron	Infron	Infron leads overall (3/5 metrics)

1. Introduction: Background, Questions, and Contributions

LLM inference–platform behavior is shaped by provider routing, prompt caching, streaming response handling, cost attribution, and fallback policy. This report treats the platform as an observable system and measures speed, cost, cache reuse, and first–token experience through strict A/B pairs.

1.1 Research Hypotheses

Hypothesis	Statement	Validation metric
H1	Stronger provider/cache affinity improves token–level cache hit rate for repeated stable prefixes.	Second–request cache–read tokens and token cache hit rate
H2	Higher cache hit rate can reduce observed cost, but does not necessarily reduce TTFT or E2E latency.	Observed cost, average TTFT, average request latency
H3	Different routing sorts change provider selection and produce different cost/throughput/latency frontiers.	Provider distribution, throughput, latency, cost

1.2 Contributions

- Uses response–returned `usage.prompt_tokens` as the input–token control while allowing small cross–platform accounting variance up to 50 tokens.
- Extends prompt–caching evaluation to cost, throughput, E2E latency, TTFT, provider distribution, reasoning telemetry, and paired statistical tests.
- Scopes every claim to observable response telemetry instead of private routing internals.

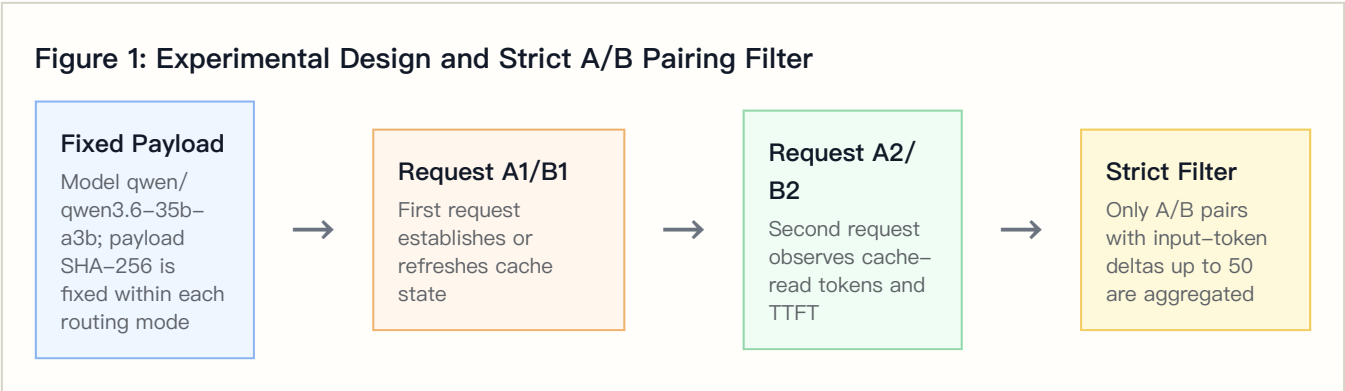
2. Experimental Design, Dataset, and Controls

2.1 Dataset Construction

The dataset is `business_representative`, covering 4 routing sorts, 2 platforms, 4 groups, and 50 rounds per group. Each round sends identical first/second Chat Completions requests; the second request observes cache-read tokens, TTFT, and E2E latency.

The built-in business templates represent stable long-context workloads such as RAG support, agent tool instructions, marketing automation, and code review.

2.2 Controlled Variables



Controlled-variable rule: within the same `sort/group/round`, both platforms must have first/second `usage.prompt_tokens` deltas no greater than 50 tokens. Total Input Tokens are computed from response-returned usage, not local tokenizer estimates.

2.3 Metric Definitions

Metric	Definition	Direction
Call cache hit rate	Share of second requests with <code>cache_read_tokens > 0</code>	Higher is better
Token cache hit rate	Second-request cache-read tokens / second-request prompt tokens	Higher is better
Observed cost	Sum of first + second request usage/cost values	Lower is better
Throughput	Completion tokens / request latency seconds	Higher is better
E2E latency	Full response latency per request	Lower is better
TTFT	Streaming first-token arrival time	Lower is better
Reasoning telemetry	Reasoning tokens from response usage	Used to explain latency, throughput, and cost

3. Experimental Environment and Data Quality

Item	Configuration
Model	qwen/qwen3.6-35b-a3b
Provider model IDs	infron: qwen/qwen3.6-35b-a3b ; openrouter: qwen/qwen3.6-35b-a3b
Platforms	Infron and OpenRouter
API protocol	/v1/chat/completions
Routing modes	Throughput First, Price First, Latency First, TTFT First
Groups	4
Rounds per group	50
Workers	24
Request mode	Streaming Chat Completions with TTFT collection
Reasoning / thinking control	No explicit reasoning/thinking parameter; model and platform defaults are preserved
Prompt length tiers	short ≈1500, medium ≈8000, long ≈32000
Excluded records	44

4. Results: Overall Metrics and Main Findings

Routing mode	Platform	Strict pairs	Total Input Tokens	Token cache hit rate	Observed cost	Throughput	E2E latency	Streaming TTFT
Throughput First	Infron	183	6518330	0.00%	\$1.10165200	1.82 tok/s	7012.66 ms	6295.91 ms
Throughput First	OpenRouter	183	6517598	0.00%	\$1.53220679	110.28 tok/s	12607.91 ms	4471.16 ms
Price First	Infron	198	6912806	0.00%	\$1.04198600	5.05 tok/s	2550.52 ms	2154.21 ms
Price First	OpenRouter	198	6912014	0.00%	\$0.97401796	5.47 tok/s	2923.52 ms	2462.44 ms
Latency First	Infron	199	6995010	0.00%	\$1.05411900	4.93 tok/s	2601.72 ms	2207.81 ms
Latency First	OpenRouter	199	6994214	0.00%	\$0.98555796	5.45 tok/s	2934.25 ms	2506.11 ms
TTFT First	Infron	198	7052586	0.00%	\$1.06273300	4.81 tok/s	2668.04 ms	2245.36 ms
TTFT First	OpenRouter	198	7051794	0.00%	\$0.99358716	4.65 tok/s	3439.11 ms	2824.57 ms

4.1 Tail Latency and Statistical Tests

Tail percentiles expose risk that averages hide.

Routing mode	Platform	P50 latency	P95 latency	P99 latency	P50 TTFT	P95 TTFT	P99 TTFT
Throughput First	Infron	3225.24 ms	36021.12 ms	76341.10 ms	2607.03 ms	32788.62 ms	75861.00 ms
Throughput First	OpenRouter	8180.42 ms	24318.35 ms	69734.35 ms	3561.57 ms	10965.10 ms	18717.57 ms
Price First	Infron	2421.66 ms	3823.41 ms	5155.24 ms	2046.43 ms	3363.70 ms	4954.53 ms
Price First	OpenRouter	2754.80 ms	4687.15 ms	7454.65 ms	2264.44 ms	3915.50 ms	7060.60 ms
Latency First	Infron	2458.84 ms	4075.08 ms	6160.28 ms	2053.23 ms	3603.98 ms	5056.58 ms
Latency First	OpenRouter	2650.83 ms	4688.94 ms	6982.76 ms	2268.77 ms	3874.55 ms	5812.22 ms
TTFT First	Infron	2508.98 ms	4255.28 ms	5551.96 ms	2132.70 ms	3484.38 ms	4838.64 ms
TTFT First	OpenRouter	2985.83 ms	6412.05 ms	10396.74 ms	2486.73 ms	5304.79 ms	8646.82 ms

Mean deltas use bootstrap 95% CIs; p-values use paired sign-flip permutation tests.

Routing mode	Metric	Mean delta	95% CI	p-value	Pairs	Interpretation
Throughput First	Latency: OpenRouter – Infron	11190.50 ms	5095.00 ms to 18202.66 ms	<0.001	183	Positive means lower Infron latency
Throughput First	TTFT: OpenRouter – Infron	–3649.50 ms	–6439.99 ms to –1011.78 ms	0.0130	183	Positive means lower Infron TTFT
Throughput First	Throughput: Infron – OpenRouter	–61.3866 tok/s	–69.0973 tok/s to –54.0288 tok/s	<0.001	183	Positive means higher Infron throughput
Throughput First	Cost: OpenRouter – Infron	\$0.00235276	\$0.00104649 to \$0.00405145	<0.001	183	Positive means lower Infron cost
Throughput First	Token Cache Hit: Infron – OpenRouter	0.00 pp	0.00 pp to 0.00 pp	1.0000	183	Positive means higher Infron cache hit
Price First	Latency: OpenRouter – Infron	745.99 ms	509.43 ms to 980.71 ms	<0.001	198	Positive means lower Infron latency
Price First	TTFT: OpenRouter – Infron	616.47 ms	407.90 ms to 827.36 ms	<0.001	198	Positive means lower Infron TTFT
Price First	Throughput: Infron – OpenRouter	–0.6455 tok/s	–0.9905 tok/s to –0.2833 tok/s	<0.001	198	Positive means higher Infron throughput
Price First	Cost: OpenRouter – Infron	\$–0.00034327	\$–0.00038772 to \$–0.00029569	<0.001	198	Positive means lower Infron cost
Price First	Token Cache Hit: Infron – OpenRouter	0.00 pp	0.00 pp to 0.00 pp	1.0000	198	Positive means higher Infron cache hit
Latency First	Latency: OpenRouter – Infron	665.05 ms	286.41 ms to 1140.86 ms	<0.001	199	Positive means lower Infron latency
Latency First	TTFT: OpenRouter – Infron	596.61 ms	242.28 ms to 1040.33 ms	<0.001	199	Positive means lower Infron TTFT

Routing mode	Metric	Mean delta	95% CI	p-value	Pairs	Interpretation
Latency First	Throughput: Infron – OpenRouter	–0.8180 tok/s	–1.1561 tok/s to –0.4885 tok/s	<0.001	199	Positive means higher Infron throughput
Latency First	Cost: OpenRouter – Infron	\$–0.00034453	\$–0.00039038 to \$–0.00029616	<0.001	199	Positive means lower Infron cost
Latency First	Token Cache Hit: Infron – OpenRouter	0.00 pp	0.00 pp to 0.00 pp	1.0000	199	Positive means higher Infron cache hit
TTFT First	Latency: OpenRouter – Infron	1542.14 ms	999.45 ms to 2239.81 ms	<0.001	198	Positive means lower Infron latency
TTFT First	TTFT: OpenRouter – Infron	1158.43 ms	687.84 ms to 1810.62 ms	<0.001	198	Positive means lower Infron TTFT
TTFT First	Throughput: Infron – OpenRouter	–0.2542 tok/s	–0.6401 tok/s to 0.1419 tok/s	0.1987	198	Positive means higher Infron throughput
TTFT First	Cost: OpenRouter – Infron	\$–0.00034922	\$–0.00039530 to \$–0.00030316	<0.001	198	Positive means lower Infron cost
TTFT First	Token Cache Hit: Infron – OpenRouter	0.00 pp	0.00 pp to 0.00 pp	1.0000	198	Positive means higher Infron cache hit

4.2 Reasoning / Thinking Control Check

This run does not explicitly set reasoning/thinking parameters and keeps model/platform defaults; this table records reasoning telemetry under default behavior.

Routing mode	Platform	Reasoning tokens	Avg reasoning tokens/request	Reasoning requests	Avg first reasoning token	Avg TTFT	Avg E2E latency
Throughput First	Infron	0	0.0000	0	0.00 ms	6295.91 ms	7012.66 ms
Throughput First	OpenRouter	497478	1359.2295	366	4471.16 ms	4471.16 ms	12607.91 ms
Price First	Infron	0	0.0000	0	0.00 ms	2154.21 ms	2550.52 ms
Price First	OpenRouter	6508	16.4343	396	2462.44 ms	2462.44 ms	2923.52 ms
Latency First	Infron	0	0.0000	0	0.00 ms	2207.81 ms	2601.72 ms
Latency First	OpenRouter	6625	16.6457	398	2506.11 ms	2506.11 ms	2934.25 ms
TTFT First	Infron	0	0.0000	0	0.00 ms	2245.36 ms	2668.04 ms
TTFT First	OpenRouter	6492	16.3939	396	2824.57 ms	2824.57 ms	3439.11 ms

4.3 API Protocol Compatibility Matrix

This run uses `/v1/chat/completions` ; this table records success response, usage, cost, and cache telemetry coverage for both platforms under that protocol.

API protocol	Endpoint	Platform	Pairs	Requests	Success rate	Usage coverage	Token usage coverage	Cost coverage	Cache telemetry coverage
chat_completions	/v1/chat/completions	Infron	800	1600	99.81%	100.00%	100.00%	100.00%	100.00%
chat_completions	/v1/chat/completions	OpenRouter	800	1600	98.81%	100.00%	100.00%	100.00%	100.00%

5. Result Visualizations by Routing Mode

The following charts are driven by the strict-paired summary data and present routing-mode, platform, cost, cache, E2E latency, Streaming TTFT, and upstream provider differences.

5.1 Core Metric Chart Overview

The charts below are generated from the same strict-paired summary data.

**Figure 3-E2:
Normalized
capability
contour**

Five metrics are normalized to 0-100, where higher is better.

**Figure 3-E3:
Cost-cache
efficiency plane**

X-axis is token cache hit rate; Y-axis is observed total cost.

**Figure 3-E1:
Routing-mode
core
metric
comparison**

Infron and OpenRouter are shown side by side by routing mode.

Figure 3–E4: E2E latency and Streaming TTFT

Solid lines show E2E latency; dashed lines show Streaming TTFT.

Figure 3–E5: Upstream provider distribution

Provider shares explain cache–domain and performance differences.

Routing–Mode Drill–Down Charts

The horizontal axis shows relative advantage: blue to the right indicates Infron advantage, orange to the left indicates OpenRouter advantage.

Figure 4–E1: Throughput First core metric comparison

Direction and magnitude of relative advantage across five metrics.

Figure 4–E2: Price First core metric comparison

Direction and magnitude of relative advantage across five metrics.

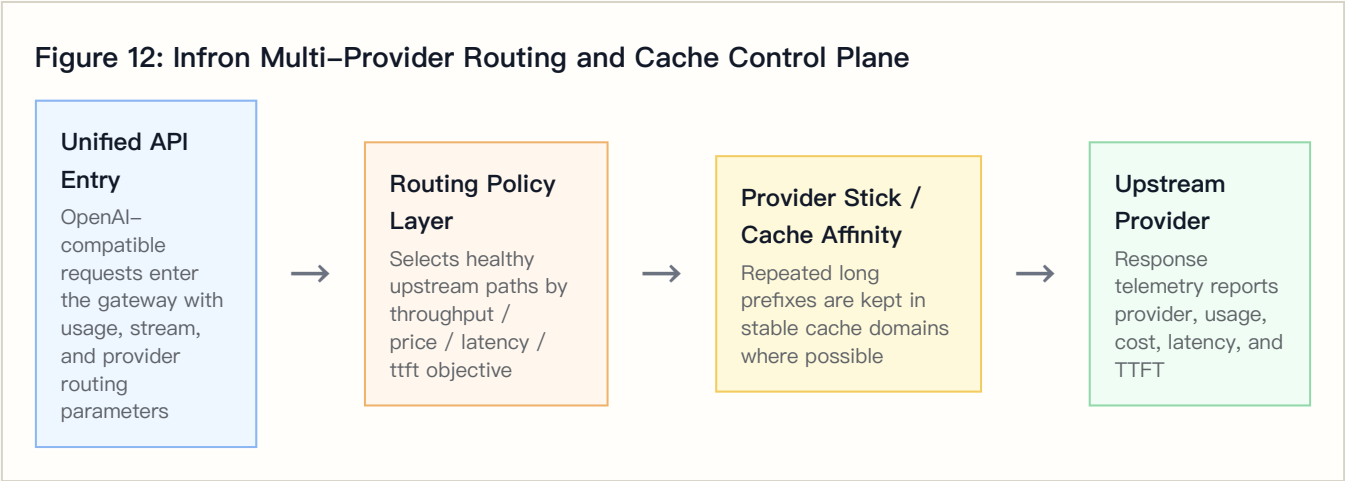
Figure 4–E3: Latency First core metric comparison

Direction and magnitude of relative advantage across five metrics.

Figure 4–E4: TTFT First core metric comparison

Direction and magnitude of relative advantage across five metrics.

6. Infron Technical Architecture and Cache/Cost Mechanism



6.1 Multi-Provider Routing and Observable Control Plane

Requests enter a unified API, and the routing layer selects healthy upstream paths according to throughput, price, latency, or ttft objectives. Whether stable long prefixes stay in the same cache domain directly affects second-request cache-read tokens.

6.2 Provider Stick and Cache Affinity

Provider stick is cache affinity, not a permanent provider lock. Its goal is to reduce cache-domain fragmentation while respecting health and SLA constraints.

6.3 Cost-Control Path

Mechanism	Cache-rate effect	Cost effect	Observable signal
Stable prefix detection	Repeated prefixes are more likely to hit cache	Reduces repeated prefill cost	Payload SHA-256 and second-request cache-read tokens
Provider stick / cache affinity	Reduces cross-domain cache fragmentation	Avoids repeated cache warm-up	Provider distribution and token cache hit rate
Health checks and fallback	Protects availability while sometimes sacrificing cache	Reduces failure cost	HTTP status, provider distribution, tail latency
Cost-aware routing	Prefers lower-cost paths under constraints	Reduces total and per-round cost	Observed cost, cost breakdown coverage, cache-read tokens

7. Provider/Route Drill-Down

Routing mode	Platform	Total requests	Attributed requests	Provider distribution
Throughput First	Infron	366	366	deepinfra 250, wafer 69, alibaba/cn 47
Throughput First	OpenRouter	366	366	AtlasCloud 333, AkashML 33
Price First	Infron	396	396	deepinfra 395, alibaba/cn 1

Routing mode	Platform	Total requests	Attributed requests	Provider distribution
Price First	OpenRouter	396	396	AkashML 396
Latency First	Infron	398	398	deepinfra 398
Latency First	OpenRouter	398	398	AkashML 398
TTFT First	Infron	396	396	deepinfra 396
TTFT First	OpenRouter	396	396	AkashML 396

Upstream Provider Detail Distribution

Routing mode	Platform	Upstream provider	Requests	Share	first/second	Covered rounds	Avg TTFT	Avg latency	Prompt tokens	Completion tokens
Throughput First	Infron	deepinfra	250	68.31%	120/130	130	2681.55 ms	3414.00 ms	4568041	3202
Throughput First	Infron	wafer	69	18.85%	32/37	41	3479.64 ms	3834.30 ms	1245543	894
Throughput First	Infron	alibaba/cn	47	12.84%	31/16	31	29655.76 ms	30820.56 ms	704746	570
Throughput First	OpenRouter	AtlasCloud	333	90.98%	166/167	174	4507.65 ms	13411.27 ms	6061381	508380
Throughput First	OpenRouter	AkashML	33	9.02%	17/16	24	4102.93 ms	4501.19 ms	456217	528
Price First	Infron	deepinfra	395	99.75%	198/197	198	2133.21 ms	2529.38 ms	6910808	5084
Price First	Infron	alibaba/cn	1	0.25%	0/1	1	10447.95 ms	10900.24 ms	1998	16
Price First	OpenRouter	AkashML	396	100.00%	198/198	198	2462.44 ms	2923.52 ms	6912014	6336
Latency First	Infron	deepinfra	398	100.00%	199/199	199	2207.81 ms	2601.72 ms	6995010	5107
Latency First	OpenRouter	AkashML	398	100.00%	199/199	199	2506.11 ms	2934.25 ms	6994214	6368
TTFT First	Infron	deepinfra	396	100.00%	198/198	198	2245.36 ms	2668.04 ms	7052586	5083
TTFT First	OpenRouter	AkashML	396	100.00%	198/198	198	2824.57 ms	3439.11 ms	7051794	6336

7.1 Cache–Rate and Cost Divergence Drill–Down

This table combines cache, cost, provider distribution, and reasoning telemetry to explain routing–mode differences.

Routing mode	Cache-hit delta	Infron cost multiple	Infron top path	OpenRouter top path	Reasoning token delta	Main attribution
Throughput First	+0.00 pp	0.72x	deepinfra 68.31%	AtlasCloud 90.98%	−497478	Cache is tied; Infron leads cost
Price First	+0.00 pp	1.07x	deepinfra 99.75%	AkashML 100.00%	−6508	Infron has higher cache but higher cost; inspect upstream unit price and completion/ reasoning tokens
Latency First	+0.00 pp	1.07x	deepinfra 100.00%	AkashML 100.00%	−6625	Infron has higher cache but higher cost; inspect upstream unit price and completion/ reasoning tokens
TTFT First	+0.00 pp	1.07x	deepinfra 100.00%	AkashML 100.00%	−6492	Infron has higher cache but higher cost; inspect upstream unit price and completion/ reasoning tokens

8. Stratified Results: Prompt–Length Cache Performance

This section aggregates second-request cache-read tokens, token-level cache hit rate, observed cost, E2E latency, and Streaming TTFT by prompt-length tier. Bold cells mark the advantaged side within each tier.

Prompt–Length Tier Overview

Prompt length tier	Target tokens	Platform	Pairs	Second prompt tokens	Second cache read tokens	Token cache hit rate	Observed cost	Avg E2E latency	Avg TTFT
short	1500	Infron	261	520226	0	0.00%	\$0.16818700	2626.43 ms	2150.90 ms
short	1500	OpenRouter	261	519704	0	0.00%	\$0.23414636	4128.76 ms	2135.23 ms
medium	8000	Infron	261	2695686	0	0.00%	\$0.84264700	3585.30 ms	3134.25 ms
medium	8000	OpenRouter	261	2695164	0	0.00%	\$0.86707414	4570.77 ms	2867.67 ms
long	32000	Infron	256	10523454	0	0.00%	\$3.24965600	4738.56 ms	4231.23 ms
long	32000	OpenRouter	256	10522942	0	0.00%	\$3.38414936	7345.25 ms	4132.85 ms

Prompt Length x Routing Mode Cache Hit Rate

Prompt length tier	Routing mode	Infron	OpenRouter	Winner
short	Throughput First	0.00%	0.00%	tie
short	Price First	0.00%	0.00%	tie
short	Latency First	0.00%	0.00%	tie
short	TTFT First	0.00%	0.00%	tie
medium	Throughput First	0.00%	0.00%	tie
medium	Price First	0.00%	0.00%	tie
medium	Latency First	0.00%	0.00%	tie
medium	TTFT First	0.00%	0.00%	tie
long	Throughput First	0.00%	0.00%	tie
long	Price First	0.00%	0.00%	tie
long	Latency First	0.00%	0.00%	tie
long	TTFT First	0.00%	0.00%	tie

9. Stratified Results: Group–Level Stability

Throughput First

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
Infron	1	47	47	0.00%	\$0.35669800	73376.27 ms	72495.67 ms
Infron	2	40	40	0.00%	\$0.24394300	6182.79 ms	5152.62 ms
Infron	3	47	47	0.00%	\$0.25199500	4571.83 ms	3892.25 ms
Infron	4	49	49	0.00%	\$0.24901600	8015.77 ms	7220.52 ms
OpenRouter	1	47	47	0.00%	\$0.35412248	22642.47 ms	6602.13 ms
OpenRouter	2	40	40	0.00%	\$0.37283790	49693.37 ms	15145.92 ms
OpenRouter	3	47	47	0.00%	\$0.39045837	23305.84 ms	14300.73 ms
OpenRouter	4	49	49	0.00%	\$0.41478804	21569.67 ms	8358.01 ms

Price First

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
Infron	1	50	50	0.00%	\$0.26138200	4012.04 ms	3623.15 ms
Infron	2	49	49	0.00%	\$0.26077100	3735.29 ms	3180.65 ms
Infron	3	49	49	0.00%	\$0.25824800	3307.70 ms	3015.72 ms

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
Infron	4	50	50	0.00%	\$0.26158500	4484.52 ms	3658.20 ms
OpenRouter	1	50	50	0.00%	\$0.24438268	4842.60 ms	3906.73 ms
OpenRouter	2	49	49	0.00%	\$0.24379264	5080.44 ms	3800.59 ms
OpenRouter	3	49	49	0.00%	\$0.24145996	4072.86 ms	3655.39 ms
OpenRouter	4	50	50	0.00%	\$0.24438268	4691.89 ms	4065.72 ms

Latency First

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
Infron	1	50	50	0.00%	\$0.26137700	3842.13 ms	3463.32 ms
Infron	2	50	50	0.00%	\$0.27311400	3785.03 ms	3108.52 ms
Infron	3	49	49	0.00%	\$0.25824800	4396.24 ms	3999.43 ms
Infron	4	50	50	0.00%	\$0.26138000	4441.58 ms	4002.67 ms
OpenRouter	1	50	50	0.00%	\$0.24438268	3945.06 ms	3576.85 ms
OpenRouter	2	50	50	0.00%	\$0.25533460	5305.07 ms	4201.66 ms
OpenRouter	3	49	49	0.00%	\$0.24145800	4291.32 ms	3744.03 ms
OpenRouter	4	50	50	0.00%	\$0.24438268	4492.70 ms	4136.66 ms

TTFT First

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
Infron	1	50	50	0.00%	\$0.26138100	4463.60 ms	3702.31 ms
Infron	2	49	49	0.00%	\$0.26998800	3683.86 ms	3276.06 ms
Infron	3	49	49	0.00%	\$0.26998300	4366.89 ms	3803.62 ms
Infron	4	50	50	0.00%	\$0.26138100	4235.40 ms	3382.72 ms
OpenRouter	1	50	50	0.00%	\$0.24438268	6985.50 ms	5304.65 ms
OpenRouter	2	49	49	0.00%	\$0.25241188	5527.35 ms	4761.40 ms
OpenRouter	3	49	49	0.00%	\$0.25240992	5720.04 ms	4630.07 ms
OpenRouter	4	50	50	0.00%	\$0.24438268	6419.30 ms	4805.62 ms

10. Discussion: Business Value, Boundaries, and Engineering Implications

Business decisions should not rely on one metric. Stable long-context and high-frequency template workloads should prioritize cache rate and cost; realtime interaction must constrain TTFT and E2E latency; batch processing often prioritizes throughput and failure cost.

Routing mode	Business objective	Observed result	Scenarios	Caveat
Throughput First	Maximize output capacity per unit time	Mixed result	Batch generation, offline summaries, backend processing	Good for throughput–first tasks; constrain cost and cache separately
Price First	Minimize request and token cost	Mixed result	High–frequency templates, support automation, marketing, RAG prefixes	Good for throughput–first tasks; constrain cost and cache separately
Latency First	Minimize full–response waiting time	Mixed result	Online chat, agent chains, IDE/ writing assistants, realtime tools	Good for throughput–first tasks; constrain cost and cache separately
TTFT First	Minimize streaming first–token time	Infron leads overall (3/5 metrics)	Streaming chat, realtime copilots, first–screen feedback	Decide with budget, SLA, and cache stability together

11. Conclusion

Routing–Mode Conclusions

Routing mode	Objective	Cache winner	Cost winner	Throughput winner	E2E latency winner	TTFT winner	Interpretation
Throughput First	Maximize output capacity per unit time	Tie	Infron	OpenRouter	Infron	OpenRouter	Mixed result
Price First	Minimize request and token cost	Tie	OpenRouter	OpenRouter	Infron	Infron	Mixed result
Latency First	Minimize full–response waiting time	Tie	OpenRouter	OpenRouter	Infron	Infron	Mixed result
TTFT First	Minimize streaming first–token time	Tie	OpenRouter	Infron	Infron	Infron	Infron leads overall (3/5 metrics)

12. Limitations, Missing Data, and Next Experiments

Limitation	Impact	Next step	Current handling
Full routing trace	Cannot prove every provider choice, fallback, and retry path hop by hop	Add provider routing trace, decision logs, and fallback reasons	Use only returned provider fields and provider distribution
Longer time window	4x50 observes short–window stability but not day–level drift	Add soak tests and repeated windows	Scope conclusions to this run
Production corpus	Built–in templates do not cover every workload distribution	Use sanitized production–stratified corpora	Discuss representative long–context templates only

Limitation	Impact	Next step	Current handling
Cost-field consistency	Cost coverage and semantics may differ by platform	Reconcile with billing and provider cost breakdown	Use only explicitly returned cost fields

13. Reproducibility Appendix

Artifact	Path
Summary	export/qwen36_35b_a3b_all_experiments/ routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions summary.json
Paired dataset	export/qwen36_35b_a3b_all_experiments/ routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions benchmark_pairs.csv
Request- level dataset	export/qwen36_35b_a3b_all_experiments/ routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions benchmark_requests.jsonl
Filtered structured records	export/qwen36_35b_a3b_all_experiments/ routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions records.json
Excluded- record audit	export/qwen36_35b_a3b_all_experiments/ routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions records_excluded.json
Test source	tests/test_rerun_routing_sort_cache_cost_ab.py
Benchmark runner source	scripts/rerun_routing_sort_cache_cost_ab.py
HTML report renderer source	scripts/render_glm52_deepseek_style_report.py
Dataset reference	business_representative built-in representative business templates; request-level export is benchmark_requests